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缩写说明

CVPR	Conference on Computer Vision and Pattern Recognition	AAAI	AAAI Conference on Artificial Intelligence
NeurIPS	Conference on Neural Information Processing Systems	TPAMI	IEEE Transactions on Pattern Analysis and Machine Intelligence
ICLR	International Conference on Learning Representations	TIP	IEEE Transactions on Image Processing
ACL	Annual Meeting of the Association for Computational Linguistics	IJCV	International Journal of Computer Vision
MM	ACM International Conference on Multimedia	TSG	IEEE Transactions on Smart Grid
ICCV	IEEE International Conference on Computer Vision		

研究方向

基础模型，研究多模态推理、智能体与情感智能、可控高效的扩散生成，以及可靠适配
鲁棒学习与优化原理，为基础模型的改进提供理论基础

工作经历

教授，中国科学技术大学，中国 2026.03 – 至今
研究科学家南洋理工大学，新加坡 2025.03 – 2026.02
研究员，商汤科技，北京 2019.07 – 2020.11

教育经历

计算机科学博士，南洋理工大学，新加坡 2021.01 – 2025.01
导师：张含望教授
电气工程硕士，清华大学，北京 2016.09 – 2019.07
电气工程学士，清华大学，北京 2012.09 – 2016.07

代表性论文

按时间顺序排列。 *：共同贡献。 †：通讯作者。

第一作者论文

IJCV 2026
1. Generalized logit adjustment: Improved fine-tuning by mitigating label bias in zero-shot vision models Imbalanced Learning.
B. Zhu, *Q. Sun, X. Yang and H. Zhang*

TPAMI 2026
2. Parallel diffusion solver via residual dirichlet policy optimization.
R. Wang, Z. Li*,* ***B. Zhu****, L. Yuan, H. Zhang, X. Yang, X. Chang and C. Zhang*

ICLR 2026
3. Reducing class-wise performance disparity via margin regularization.
B. Zhu, *K. Zhao, J. Cui, Q. Sun, Y. Zhou, X. Yang and H. Zhang*

NeurIPS 2025
4. Adaptive stochastic coefficients for accelerating diffusion sampling.
R. Wang,* ***B. Zhu****, J. Li, L. Yuan and C. Zhang*

ICCV 2025
5. Distilling parallel gradients for fast ODE solvers of diffusion models.
B. Zhu*, R. Wang*, T. Zhao, H. Zhang and C. Zhang*

CVPR 2025
Highlight（3.0%）
6. Project-probe-aggregate: efficient fine-tuning for group robustness.
B. Zhu, *J. Cui, H. Zhang and C. Zhang*

NeurIPS 2024
7. Robust fine-tuning of zero-shot models via variance reduction.
B. Zhu, *J. Cui and H. Zhang*

NeurIPS 2024
Spotlight（2.1%）
8. Enhancing zero-shot vision models by label-free prompt distribution learning and bias correcting.
X. Zhu,* ***B. Zhu****, Y. Tan, S. Wang, Y. Hao and H. Zhang*

MM 2024
Oral（4.0%）
9. Selective vision-Language subspace projection for few-shot CLIP.
X. Zhu,* ***B. Zhu****, Y. Tan, S. Wang, Y. Hao and H. Zhang*

NeurIPS 2023
10. Generalized logit adjustment: Calibrating fine-tuned models by removing label bias in foundation models.
B. Zhu, *K. Tang, Q. Sun and H. Zhang*

ICCV 2023
11. Prompt-aligned gradient for prompt tuning.
B. Zhu, *Y. Niu, Y. Han, Y. Wu and H. Zhang*

AAAI 2023
Oral（10.8%）
12. Debiased fine-tuning for vision-language models by prompt regularization.
B. Zhu, *Y. Niu, S. Lee, M. Hur and H. Zhang*

AAAI 2022
Oral（4.2%）
13. Cross-domain empirical risk minimization for unbiased long-tailed classification.
B. Zhu, *Y. Niu, X. Hua and H. Zhang*

TIP 2021
14. Structure-coherent deep feature learning for robust face alignment.
C. Lin,* ***B. Zhu****, Q. Wang, R. Liao, C. Qian, J. Lu and J. Zhou*

通讯作者论文

NeurIPS 2026
Oral（0.4%）
15. Thinking with images as continuous actions: Numerical visual chain-of-thought.
*K. Zhao, †**B. Zhu**, J. Zhou, X. Zhu, Z. Yue and H. Zhang*

MM 2026
Oral
16. Layer roles matter: Mitigating modality imbalance in multimodal emotion reasoning via adaptive layer profiling.
*P. Huang, P. Song, †**B. Zhu**, Z. Han, J. Hu and X. Yang*

ICML 2026
17. Omni-perception policy optimization for multimodal emotion reasoning.
*Z. Han, †**B. Zhu**, W. Tong, P. Shao, P. Song, X. Wang, J. Chen, L. Lu and X. Yang*

CVPR 2026
18. Adapting point cloud analysis via multimodal Bayesian distribution learning.
*X. Zhu, L. Yi, S. Wang, W. Zhu, Y. Wu, †**B. Zhu** and H. Zhang*

TIP 2026
19. Hybrid granularity distribution estimation for few-shot learning: statistics transfer from categories and instances.
*S. Wang, T. Qi, X. Zhu, Y. Hao, †**B. Zhu**, H. Zhang and M. Wang*

ICLR 2026
20. Real-time motion-controllable autoregressive video diffusion.
*K. Zhao, J. Shi, †**B. Zhu**, J. Zhou, X. Shen, Y. Zhou, Q. Sun and H. Zhang*

ICLR 2026
21. Look carefully: adaptive visual reinforcements in multimodal large language models for hallucination mitigation.
*X. Zhu, K. Zhao, L. Yi, S. Wang, Z. Wang, †**B. Zhu**, H. Zhang and X. He*

AAAI 2026
22. DEPO: dual-efficiency preference optimization for LLM agents.
*S. Chen, M. Zhao, L. Xu, Y. Zhao, †**B. Zhu**, H. Zhang, S. Zhao and C. Lu*

MM 2025
Oral
23. Benchmarking and bridging emotion conflicts for multimodal emotion reasoning.
*Z. Han, †**B. Zhu**, Y. Xu, P. Song and X. Yang*

ICCV 2025
24. Unsupervised visual chain-of-thought reasoning via preference optimization.
*K. Zhao, †**B. Zhu**, Q. Sun and H. Zhang*

TSG 2019
25. Fault location for radial distribution network via topology and reclosure-generating traveling waves.
*S. Shi, †**B. Zhu**, A. Lei and X. Dong*

TSG 2018
26. Fault classification for transmission lines based on group sparse representation.
*S. Shi, †**B. Zhu**, S. Mirsaeidi and X. Dong*

其他论文

NeurIPS 2026
27. Beyond SFT-to-RL: Pre-alignment via black-box on-policy distillation for multimodal RL.
*S. Wang, W. Huang, X. Yu, Z. Yang, H. Lin, K. Wu, C. Xiao, W. Wang, C. Chen, **B. Zhu**, Y. Zhang and C. Qin*

NeurIPS 2026
28. TopoRefine: Topology-aware correspondence and residual refinement for training-free subject-consistent generation.
*Z. Gao, Z. Ti, C. Zhao, **B. Zhu**, X. Hu, Y. Ge, J. Yang and Y. Tai*

TPAMI 2026
29. Generalized kullback-leibler divergence loss.
*J. Cui, **B. Zhu**, Q. Xu, Z. Tian, X. Qi, B. Yu, H. Zhang and R. Hong*

TIP 2026
30. Detail++: training-free detail enhancer for text-to-image diffusion models
*L. Chen, J. Wang, Z. Pan, **B. Zhu**, X. Yang and C. Zhang*

ACL 2026
Findings
31. AMA: adaptive memory via multi-agent collaboration.
*W. Huang, Z. Wang, H. Lin, S. Wang, B. Xu, Q. Li, **B. Zhu**, L. Yang and C. Qin*

ACL 2026
Oral（前 5%）
32. Mitigating hallucinations in large vision-language models without performance degradation.
*X. Zhu, J. Fang, S. Wang, **B. Zhu**, Z. Wang, Y. Yang and X. He*

ICML 2026
33. Robustifying vision-language models via test-time prompt adaptation.
*X. Zhu, H. Wu, S. Wang, **B. Zhu**, J. Ge, J. Zhang and L. Chen*

CVPR 2026
34. Principled steering via null-space projection for jailbreak defense in vision-language models
*X. Zhu, **B. Zhu**, S. Wang, J. Fang, K. Zhao, H. Zhang and X. He*

CVPR 2026
35. Few-step diffusion sampling through instance-aware discretizations
*L. Yuan, R. Wang, T. Zhao, D. Fu, M. Lei, **B. Zhu** and C. Zhang*

CVPR 2026
36. ACE-merging: data-free model merging with adaptive covariance estimation
*B. Xu, H. Wu, H. Lin, W. Huang, **B. Zhu**, Y. Shu and C. Qin*

ICLR 2026
37. CoDi: subject-consistent and pose-diverse text-to-image generation.
*Z. Gao, **B. Zhu**, L. Yao, J, Yang and Y. Tai*

ICLR 2026
38. GuardAlign: robust safety alignment in multimodal large language models.
*X. Zhu, **B. Zhu**, J. Fang, S. Wang, Y. Zhang, X. Wang and X. He*

ICLR 2026
39. Free lunch for stabilizing rectified flow inversion.

*C. Wang, **B. Zhu** and C. Zhang*

ICLR 2026
40. Streaming drag-oriented interactive video manipulation: drag anything, anytime!
*J. Zhou, Y. Zhou, K. Zhao, Q. Xu, **B. Zhu**, R. Hong and H. Zhang*

AAAI 2026
41. Hierarchical semantic alignment for image clustering.
*X. Zhu, **B. Zhu**, Y. Li, J. Fang, S. Wang, K. Zhao and H. Zhang*

预印本 2025
42. Generative distribution distillation.
*J. Cui, **B. Zhu**, Q. Xu, X. Xu, P. Chen, X. Qi, B. Yu, H. Zhang and R. Hong*

NeurIPS 2025
Spotlight（3.2%）
43. Enhancing CLIP robustness via cross-modality alignment.
*X. Zhu, **B. Zhu**, S. Wang, K. Zhao and H. Zhang*

ICCV 2025
44. Dynamic multimodal prototype learning in vision-language models.
*X. Zhu, S. Wang, **B. Zhu**, M. Li, Y. Li, J. Fang, Z. Wang, D. Wang and H. Zhang*

CVPR 2025
45. Devils in middle layers of large vision-language models: interpreting, detecting and mitigating object hallucinations via attention lens.
*Z. Jiang, J. Chen, **B. Zhu**, T. Luo, Y. Shen and X. Yang*

CVPR 2025
46. Stylestudio: text-driven style transfer with selective control of style elements.
*M. Lei, X. Song, **B. Zhu**, H. Wang and C. Zhang*

CVPR 2024
47. Classes are not equal: an empirical study on image recognition fairness.
*J. Cui, **B. Zhu**, X. Wen, X. Qi, B. Yu and H. Zhang*

AAAI 2023
Oral（10.8%）
48. Leveraging modality-specific representations for audio-visual speech recognition via reinforcement learning.
*C. Chen, Y. Hu, Q. Zhang, H. Zou, **B. Zhu** and E. Chng*

专利

中国专利
CN122433924A
1. 基于快慢思考协同的多模态情感推理方法和系统
杨勋；韩知渊；张嘉宇；**朱贝尔**；宋培培

中国专利
CN122414419A
2. 基于强化学习与全模态感知优化的情感推理方法及系统
杨勋；韩知渊；张嘉宇；**朱贝尔**；宋培培

美国专利
US11308710B2
3. Polygonal Region Detection.
Zhu Beier、*Zhang Rui*

中国专利
CN110136156A
4. 一种多边形区域检测方法及装置
朱贝尔；张睿

学术服务

期刊审稿人： TPAMI, TIP, TMM, TCSVT, TCE, IJCV, Information Fusion

会议审稿人： CVPR, ICCV, AAAI, MM, UAI, AISTATS, NeurIPS（2025 年**优秀审稿人**），ICLR, ICML（2026 年**优秀审稿人**），ACL, KDD。

荣誉奖项

最佳论文，国际工程智能大会，中国	2026
荣誉奖，Nanyang Speech Forum，新加坡	2023
AISG PhD Fellowship，National Research Foundation，新加坡	2021
一等奖学金，清华大学，中国	2018
奖学金，国家留学基金管理委员会，中国	2014

实习经历

研究实习生，商汤科技，北京	2018.03 – 2018.10
研究实习生，腾讯优图实验室，上海	2018.07 – 2018.09
研究实习生，美团，北京	2017.08 – 2017.12

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